Mohammad Orabe

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I'm a software developer interested in data analysis, machine learning, and neurotechnology. I build software solutions to address challenges at the intersection of these fields. Portfolio available on website.

Experience

Machine Learning Research / Part-Time
Jul 2023 – Present
QAI Labs (PTB & Charité)

- **PTB – National Metrology Institute of Germany**
  - Developed an open-source toolbox for benchmarking uncertainty & calibration in brain source imaging.
  - Implemented an automated framework to evaluate the reliability of inverse machine learning methods—including data generation, performance evaluation, and visualization.
- **Charité – Universitätsmedizin Berlin**
  - Contributed methods for analyzing brain connectivity (how different regions interact) to a widely used open-source software package.
  - Implemented advanced methods for electrophysiological data analysis and preprocessing.

Data Science
Student Research Assistent
Jul 2021 – Jun 2023
Winter Lab

Implemented machine learning pipelines and statistical toolboxes for different tracking environments. Applied deep learning and computer vision techniques for object detection and developed reproducible analytics workflows.

Technical Support
Feb 2017 – Jul 2020
IIK Berlin

Provided IT support and maintained Linux/Windows systems. Assisted with system troubleshooting and automated routine administrative tasks.

Education

M.Sc. in Computer Science
Oct 2022 – Sep 2025
TU Berlin

Focused on machine learning, data science, and intelligent systems, with coursework covering deep learning, probabilistic modeling, and software engineering. Thesis: Ongoing

B.Sc. in Computer Science
Oct 2018 – Sep 2022
TU Berlin

Core curriculum included algorithms, programming, computer architecture, and applied machine learning. Thesis: Pose Estimation and Multi-Object Tracking with Deep Learning.

Internships

Software Development
Jul 2020 – Oct 2020
DAI Lab

Contributed to the development of distributed eHealth and agent-based systems. Prototyped electrical circuits for wearable sensor platforms.

Open-Source Software – Lead Developer

CaliBrain (Python)
Model Uncertainty Evaluation
2024 – Present

Designed a modular benchmarking framework to evaluate model reliability and calibration in data-driven ML inverse problems.

GaitMod (Python)
Real-Time Gait Detection Toolkit
2024 – Present

Built a library for real-time classification of impaired movement using invasive human brain signals. Integrated deep learning models for precise state prediction and adaptive feedback.

GroupMouseTrack (Python) Multi-Object Tracking 2022	Developed a deep learning tool for multi-object tracking and pose estimation—integrating sensor data with computer vision to preserve individual identities over time.
ColonyRack (R) Behavioral Data Processing Suite 2021 – 2023	Created a toolbox for processing and analyzing large-scale time-series data in behavioral studies to enabled efficient extraction of activity patterns and social interaction metrics.

Teaching & Mentorship

Founder & Lead Instructor MathCodeLab 2023 – Present	Launched a tech education platform offering in-person & online courses in coding and data science. Taught 10+ intensive 12-week courses. Built a vibrant community of over 240 learners.
Teaching Assistant Biomedical Data Science Winter 2024/2025 – TU Berlin	Assisted in a Master's-level seminar focused on advanced machine learning for biomedical data science. Supported students with project supervision, course logistics, and technical Q&A.

ML & Data Science Projects (Selected)

MNE-Python Signal Analysis	Contributed advanced methods for analyzing brain-related signals to a widely used open-source scientific library.
neurolib Whole–Brain Simulation	Extended an open-source platform to support simulations of brain activity using standardized anatomical models.
Medical Image Analysis MRI Segmentation	Trained deep learning models to detect and segment tumors in medical imaging data.
Multi-Modal VAE Generative Model	Built a generative model that learns shared patterns from images and associated labels, enabling conversions between data formats.

Technical Skills

Languages	Python, Java, R, MATLAB, C/C++
ML Libraries & Frameworks	TensorFlow, PyTorch, scikit-learn, OpenCV, Pandas, Numpy
Applied ML Domains	NLP, Computer Vision, Time Series, Signal Processing
Machine Learning Architectures & Models	CNN, RNN(LSTM); Transformers (LLMs); Generative (GANs, AEs), Reinforcement Learning
Cloud & DevOps	Docker, AWS, Azure, CI/CD
Data Engineering	SQL, ETL, Apache Kafka & Spark
Software Development Tools	Git, Bash, SSH, gdb, Unit Testing, API Design

Languages

English	Fluent
German	Professional

References

Available upon request.